INSPECTO V3 — USER GUIDE

Pocket Dual-Channel Universal Industrial I/O Tester & Calibrator

USB-C POWERED
0–60V IN / 0–22V OUT

1. Meet Inspecto V3: Your Pocket Industrial Lab

OVERVIEW

Inspecto V3 turns your Android smartphone into a portable industrial signal analyzer and voltage source. Powered directly from your phone's USB-C port, the board boosts 5V USB power into an internal **24V rail** and provides two independent universal channels (**CH1** and **CH2**) that switch between measuring external voltages and generating test signals.



Figure 1: Inspecto V3 draws 5V from your phone via USB-C and provides two software-configurable I/O terminals (CH1 & CH2) plus common GND.

2. Three Operating Modes in Every Channel

CAPABILITIES

- ● **Analog Input Mode (AI — 0.17V to 60V DC):** Safely measures external DC voltages from sensors, batteries, and 24V/48V industrial panels. The board automatically opens its internal optical relay to isolate the output circuit and displays live voltage and oscilloscope waveforms on your phone.
- ▲ **Analog Output Mode (AO — 0.00V to 23.00V DC):** Closes the internal relay and drives a smooth, regulated DC voltage that you set in the app (such as 5.00V, 10.00V, or 15.00V). Ideal for simulating 0–10V analog sensors, testing control valves, and calibrating gauges.
- ■ **Digital Output Mode (DO — 0V LOW / 23V HIGH):** Turns the channel into a fast 23V industrial logic switch. Tap **HIGH (1)** to output ~23V DC for triggering PLC inputs and indicator lamps, or tap **LOW (0)** to drop the line to 0.00V.

Onboard Status LED Heartbeat: Watch the LED on the Inspecto V3 board to verify terminal safety at a glance. A **slow blink (every 0.5 seconds)** indicates both channels are safely isolated in Analog Input mode. A **rapid blink (every 0.15 seconds)** warns that at least one channel is actively outputting voltage.

3. Getting Started in 4 Simple Steps

QUICK SETUP

1. **Install the Android App:** Scan your Inspecto QR code or open `github.com/ShadiElshazly/INSPECTO-APP/releases/latest/download/Inspecto.apk` on your Android phone to download and install the companion app.
2. **Connect via USB-C:** Plug the Inspecto V3 board into your phone's USB-C port. The onboard status LED will begin blinking slowly, and Android will display a USB permission prompt—tap **OK** to connect.
3. **Wire Your Terminals:** Always connect the black reference wire to **GND** first, then connect your red signal wire to **CH1** or **CH2**. Both channels always start in safe **Analog Input (AI)** mode on power-up.
4. **Select Mode & Operate:** On the app dashboard, choose **AI** to read voltage on the live gauge and oscilloscope graph, **AO** to type or slide to a target voltage (0–23V), or **DO** to toggle 23V logic power.

4. Visual Wiring & Loopback Self-Test Guide

CONNECTION DIAGRAMS

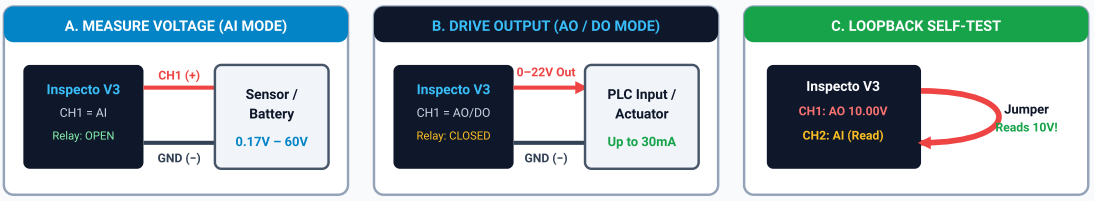


Figure 2: (A) High-impedance voltage measurement, (B) Active 0–23V signal generation, and (C) Loopback test connecting CH1 directly to CH2.

5. Self-Healing Protection & PC Serial Commands

SAFETY & ADVANCED USE

⚡ **Automatic Self-Healing Protection:** Inspecto V3 features resettable PPTC fuses (`F1`, `F2`, `U13`), reverse-polarity Schottky clamping, and 27V Zener surge guards. If a wire is accidentally shorted to ground or exposed to reverse voltage, the board safely limits fault current to **~30mA**. Simply disconnect the faulted wire for 5 seconds and the board resets itself automatically.

💻 **PC USB Serial Control (Hercules / Terminal — 115200 Baud, 8N1):**

• Read live JSON status: `D` | Set CH1 or CH2 to safe input mode: `CH1:AI` or `CH2:AI`

• Set Analog Output voltage: `CH1:AO:10.00` or `CH2:AO:5.50` | Digital 22V HIGH/LOW: `CH1:DO:1` / `CH1:DO:0`